

Summary

We assembled the genome of *Escherichia coli* O157:H7 using two different approaches: an assembly from Illumina short reads only and a hybrid assembly (Illumina reads combined with Nanopore long reads). FastQC was used to assess the quality of the Illumina reads, followed by trimming adapters and low-quality bases with Trim Galore. SPAdes was employed to assemble both assemblies and QUAST to assess the quality of the assemblies. FastQC showed improved quality of the Illumina reads after trimming. QUAST analysis revealed that the Illumina-only assembly had lower contiguity but significantly fewer misassemblies compared to the hybrid assembly, where 86% of the genome was misassembled. Due to its superior structural accuracy, the Illumina-only assembly is more suitable for downstream analysis.

Introduction

Escherichia coli O157:H7 is a highly pathogenic strain that can cause bloody diarrhea, hemolytic uremic syndrome, kidney failure, and even death [1]. Its recent outbreak in McDonald's caused 90 people to be infected and 1 death, highlighting the ongoing risk of this bacterium in the food industry and the need for effective monitoring and control strategies [2].

Genome assembly of *E. coli* O157:H7 can help us understand the genetic basis of its pathogenicity and aid in outbreak response [3]. In this study, we aim to assemble the genome of *E. coli* O157:H7 using short, paired-end reads from Illumina MiSeq and assess whether incorporating long reads generated by Oxford Nanopore Technology Minion improves the assembly's completeness and accuracy for downstream analysis.

Methods

Sequencing Data Preparation

Paired-end short reads from *E. coli* O157:H7 were obtained from Illumina MiSeq sequencer. The reads were uploaded to the University of Glasgow's Galaxy server, consisting of forward and reverse reads (TUV_S7_L001_R1_001.fastq and TUV_S7_L001_R2_001.fastq). Assessment of quality was performed using FastQC (version 0.12.1) to evaluate the raw reads.

Quality Trimming

Adapter sequences and low-quality bases were trimmed using Trim Galore (version 0.6.7), with a Phred score threshold of 20. Reads that became shorter than 100 bp after trimming were discarded. The trimmed reads were re-assessed for quality using FastQC to ensure improved quality before proceeding with assembly.

Genome Assemblies, Pre-assembly & Post-assembly Corrections

Trimmed reads were assembled using SPAdes (version 3.15.4), which includes a pre-assembly error correction step to improve the accuracy of input reads and a post-assembly mismatch correction step to refine the assembled sequences by minimizing mismatches and

other errors. The trimmed short reads were used for the first assembly that comprises of Illumina reads only.

Low-frequency k-mers are often due to sequencing errors. Pre-assembly error correction step ensures that low-frequency k-mers are removed and high-quality k-mers are kept for assembly. K-mers lengths of 21, 33, 55, 77, 99 and 127 were used to generate contigs in our study. Post-assembly error correction step utilises alignment tool, BWA to align the original reads to the assembled contigs. The consensus bases are chosen based on the most frequent bases found among the reads. If the bases in the contigs do not match the consensus bases from the aligned reads, the contigs are corrected to match the reads.

To create a second assembly, a hybrid assembly, the above-mentioned trimmed Illumina short reads were combined with long reads generated using Oxford Nanopore Technology MinION (TUV359.2d.fq). Both datasets were assembled using SPAdes.

Assembly Quality Assessment

The quality of the assemblies was evaluated using QUAST. For both the Illumina and hybrid assemblies, the contigs generated by SPAdes were compared to the reference genome (*Escherichia coli* O157:H7 str. EDL933; accession: GCA_000732965.ASM73296v1.31.dna.genome.fa). A lower threshold of 1000 bp was set for contig length to focus on significant assembly pieces.

Results

(I) Basic Statistics of All Illumina Reads

Read Quality Assessment

The forward read of the Illumina dataset contained 347,436 sequences, comprising a total of 104.2 Mbp with sequence lengths ranging from 35 to 301 bp. After trimming, the number of forward reads was reduced to 333,828 sequences (total bases=90.8 Mbp), with sequence lengths between 100 and 301 bp. The GC content remained consistent at 50% before and after trimming (Table 1).

The trimming process identified that 23.7% of the forward reads (82,304 reads) contained adapter sequences, and 9.9% of bases (10,353,880 bp) were quality-trimmed. For the reverse reads, 24.7% (85,661 reads) contained adapters, and 27.7% of bases (28,860,397 bp) were trimmed for quality (Table 2). The reverse reads showed a higher proportion of quality-trimmed bases compared to the forward reads, likely due to greater quality degradation towards the end of the reads or increased adapter presence.

The quality-based trimming process resulted in the removal of 13,608 sequence pairs (3.92%) because at least one read in the pair was shorter than the cut-off length of 100 bp. This left 333,828 forward reads and their corresponding reverse reads available for the assembly. The reduction in the number of reads and the overall trimming process ensured a high-quality dataset suitable for downstream genome assembly.

	Forward Reads	Trimmed Forward Reads
Total Sequences	347,436	333,828
Total Bases (Mbp)	104.2	90.8
Sequence Length	35-301	100-301
%GC	50	50

Table 1. Basic statistics of forward reads before and after being trimmed by Trim Galore

	Forward Reads	Reverse Reads
Total Reads Processed	347,436	347,436
Reads with Adaptors	82,304 (23.7%)	85,661 (24.7%)
Total Base Pairs Processed (bp)	104,217,539	104,281,400
Quality-trimmed (bp)	10,353,880 (9.9%)	28,860,397 (27.7%)

Table 2. Basic statistics of forward reads and reverse reads before and after being trimmed by Trim Galore

(II)Key FactQC of Untrimmed and Trimmed Reads

Per-base Sequence Quality Improvement of Illumina Forward Reads

The per-base sequence quality of the forward Illumina reads before and after trimming is shown in Figure 1. The initial quality (left panel) demonstrates a significant drop in Phred scores towards the latter part of the reads, with many positions falling below a Phred score of 20 (positions 264 and onwards have median values below a Phred score of 20, as shown by the red area in the figure), indicating reduced reliability and the presence of low-quality bases. The average quality also shows increased variability towards the end, with a wide range of interquartile values across positions.

After applying Trim Galore (right panel), the quality improved significantly, especially in the latter part of the reads. The trimmed reads show consistently higher Phred scores across all positions, and the variability has been reduced. The majority of bases now fall within the green zone, indicating high confidence. All positions in the reads have median Phred scores above 20. This demonstrates the effectiveness of the trimming process in improving the overall quality of the sequencing reads, making them more suitable for downstream analysis and genome assembly.

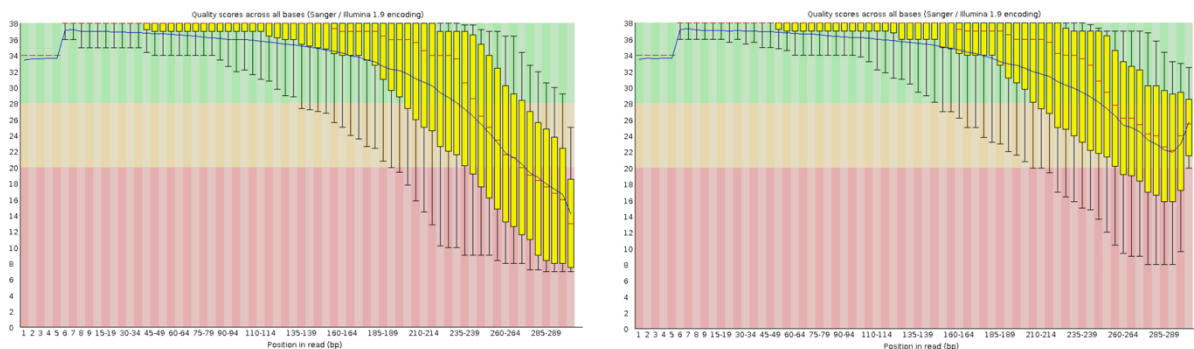


Figure 1. Per-base sequence quality of forward Illumina reads before (left) and after (right) trimming. The plots indicate Phred quality scores across each position in the reads, with the green, yellow, and red areas representing high, moderate, and low-quality scores, respectively. Higher scores indicate better base calls. The red lines represent median scores, blue lines represent mean scores and yellow boxes represent the inter-quartile ranges.

Per-base Sequence Content Before and After Trimming

Figure 2 shows the per-base sequence content of the forward Illumina reads before (left panel) and after trimming (right panel). In the untrimmed reads, the sequence content diverges towards the latter part, with a significant imbalance starting at position 210 bp of the reads. This is likely indicative of adapter contamination and low-quality regions.

After trimming with Trim Galore (right panel), the per-base sequence content becomes much more stable (the graph was stable until position 264 bp of the reads), with each nucleotide consistently represented at around 25%. This balance indicates that the trimming process successfully removed adapter sequences and improved overall sequence quality, resulting in a dataset more appropriate for downstream analysis. The elimination of extreme fluctuations, particularly towards the end of the reads, demonstrates an effective improvement in sequence uniformity.

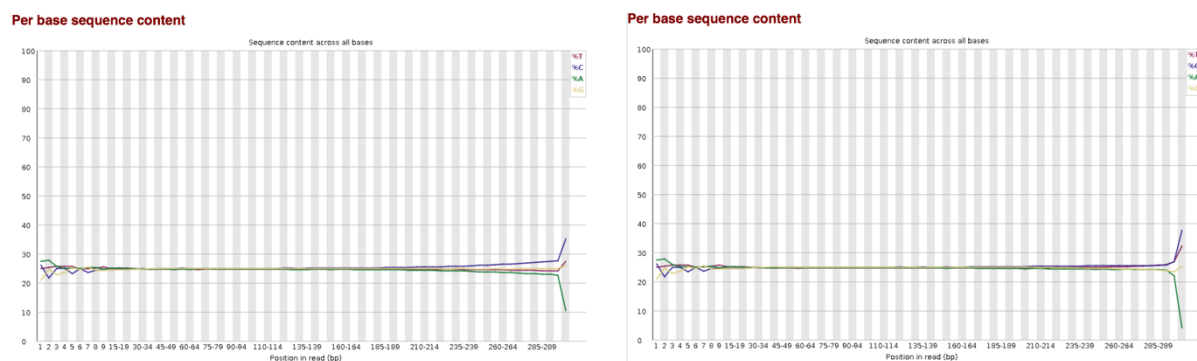


Figure 2. Per-base sequence content of forward Illumina reads before (left) and after (right) trimming. The plots show the proportion of each nucleotide (A, T, C, G) across the read positions, with trimming resulting in improved uniformity of nucleotide distribution.

(III) Comparison of Illumina Assembly & Hybrid Assembly

Illumina Assembly Evaluation with QUAST

The quality of the genome assembly was assessed using QUAST. The assembly was made up of 95 contigs (≥ 1000 bp), with a total length of 5,219,547 bp (Table 3). The largest contig measured 505,119 bp. The reference genome length was 5,639,399 bp, with a GC content of 50.43%, which closely matched the GC content of the assembled contigs at 50.28%.

The N50 and NG50 values, which indicate the contig length at which half the genome is covered, were 148,373 bp and 146,855 bp, respectively. These metrics reflect the overall quality of the assembly, with higher values indicating longer, more complete contigs. The L50 value was 9, meaning that 9 contigs were needed to cover half of the total assembly length.

The genome fraction covered by the assembled contigs was 95.103%, suggesting that the majority of the reference genome was represented in the assembly. There was only 1 misassembly detected (Figure 3), with a misassembled contig length of 98,856 bp.

Overall, the assembly metrics, including the N50, NG50, and genome fraction, demonstrate a high-quality assembly with minimal errors, making it suitable for downstream analysis.

	Illumina Assembly	Nanopore + Illumina Hybrid Assembly
Genome Fraction Covered (%)	95.103	97.003
NGA50	146,855	3,572,033
N50	148,373	3,572,033
Total Length of Contigs ($\geq$ 1k bp)	5,219,547	5,389,753
Number of Contigs ($\geq$ 1k bp)	95	12
Misassembled Contigs	1	3
Misassembled Contigs Length	98,856	4,658,887

Table 3. QUAST key summary of Illumina Assembly and Hybrid Assembly to show the completeness and accuracy of the assemblies

QUAST Evaluation of Hybrid Assembly

The hybrid assembly, incorporating both trimmed Illumina short reads and Nanopore long reads, yielded a total of 12 contigs ($\geq$ 1000 bp), with a total length of 5,389,753 bp. The largest contig measured 4,233,969 bp, representing a substantial portion of the reference genome length (5,639,399 bp). The GC content of the assembled contigs (50.40%) was consistent with that of the reference genome (50.43%).

The N50 and NG50 values for the assembly were both 4,233,969 bp, and the L50 value was 1, indicating that a single contig covered 50% of the assembly. The genome fraction covered was 97.003%, suggesting a high level of completeness.

However, 4,658,887 bp of the total contig length were identified as misassembled, representing approximately 86% of the assembly (Figure 3). This high proportion of misassemblies raises concerns about the structural accuracy of the hybrid assembly.

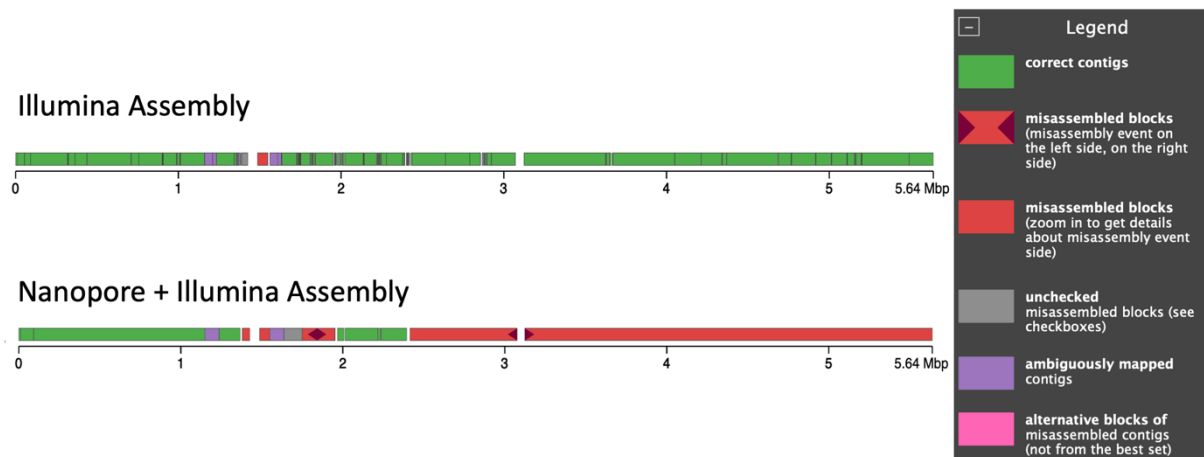


Figure 3. ICARUS QUAST contig browser view of the Illumina-only assembly (top) and the hybrid Nanopore + Illumina assembly (bottom). The green regions represent correctly assembled contigs. The red and purple regions indicate misassembled blocks or ambiguously mapped contigs. The Illumina assembly shows fewer misassembled regions, indicating greater structural accuracy.

Discussion

Explanation on How Trim Galore Improved Reads Quality

Trim Galore improved the quality of Illumina reads significantly. Trim Galore achieved it through adapter removal, quality trimming and length filtering. Sequence quality has the tendency to degrade towards the end of the reads. By trimming bases with Phred scores below the threshold (in our case, Phred score 20) and removing the contaminating adapter contents at the end of the sequences, the overall read quality improved, with no low-confidence regions presence after trimming. Reads that became shorter than 100 after trimming were discarded as very short reads tend to be associated with sequencing errors. Trim Galore ensured that only reads of sufficient length and quality were kept, contributing to an overall higher-quality dataset [4].

Illumina-only Assembly Outperformed Hybrid Assembly

Based on the QUAST metrics, the assembly using only Illumina reads appears to be superior to the hybrid assembly in terms of accuracy and reliability. Although the hybrid assembly produced larger contigs and higher contiguity (e.g., N50 and NG50 values), it also exhibited a significant proportion of misassemblies, with 86% of the total contig length affected by structural inaccuracies. This raises concerns about the quality and trustworthiness of the hybrid assembly.

In contrast, the assembly using only Illumina reads showed fewer misassemblies and a lower proportion of incorrect regions, indicating a more accurate representation of the genome. While the contiguity of the Illumina-only assembly may be lower compared to the hybrid approach, its overall higher accuracy makes it a more reliable choice for downstream applications where precision is crucial [5].

Thus, the Illumina-only assembly is preferable for applications where structural accuracy is more important than contiguity.

Future Study: Achieve Single Contig with High Accuracy

In our future study, we hope to achieve single contig with low error rate for our assemblies.

Sutton et al. (2021) have demonstrated that low coverage of Nanopore long reads produced assemblies of six to nine contigs with high rates of mismatches compared to assemblies from high coverage libraries. Likewise, the assembly containing long reads from our study shared a similar problem. Sutton et al. (2021) increased the coverages of Illumina reads to 150x and Nanopore long reads to 227x. The hybrid assembly formed using MasuRCA managed to produce single contig with low rate of mismatches, insertions and deletions [6].

PacBio has developed highly accurate long reads known as HiFi reads, providing both accuracy and completeness of the sequence [7]. A fairly new assembler DEGAP could be used on HiFi reads to close the gaps of the assembly and to increase the contiguity greatly. Huang et al. (2024) have used DEGAP to reduce nine contigs to two contigs on HiFi reads done on complex genome. The combination of HiFi and DEGAP are able to provide high accuracy and contiguity of the assembly and hopefully reaching single contig [8].

Hence, to improve on our assemblies, we can consider increasing coverage, using assembler (eg., MasuRCA) built for hybrid assemblies, switching to more accurate sequencing that produces HiFi reads and DEGAP to generate accurate and high contiguity assembly.

Conclusion

Our study has demonstrated that incorporating Nanopore long reads with Illumina reads can greatly improve contiguity, but at a big expense of accuracy. Therefore, for downstream analysis, we would choose the assembly that has Illumina reads only, due to its reliability and accuracy.

Reference

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